

Project: Human-in-the-Loop Power Tool User Monitoring

The user is a vital part of the “system” during the operation of power tools. Accounting for their condition and handling of the tool is a way to optimize safety and work efficiency. The goal of this project is to build a suite of smart watch features including but not limited to the following user stories:

1. As a trade worker I would like to see my current and daily average BPM to know how hard I’m working.
2. As a trade worker, I want to receive an alert including haptic feedback when my heart rate goes above or below a user-defined target zone so that I can adjust my pace to remain safe and productive.
3. As a trade worker I would like to see my hourly average BPM throughout the day to see when my peak work times are.
4. As a trade worker I would like the tracking of my data to happen without the screen on to maximize the battery life of the device and minimize distractions.
5. As a trade worker I would like the watch to be wearable on either hand so that I don’t have to worry about which hand I’m using for my job.
6. As a trade worker I would like to track active duration of high impact (HR) activities so that I can know how long my bursts of high impact activities are.
7. As a trade worker I would like the option to track the distance and average speed traveled for my activity so I can know how far I traveled in one workday.
8. As a trade worker, I want the tracking to start automatically when I put the watch on so that I do not need to remember to turn it on or off.
9. As a trade worker, I want to see my key performance metrics at a glance during my workday so that I can monitor my effort.
10. As a team manager, I want to be able to set a break reminder cadence based on time to send a haptic reminder to team members to take a short break in order to avoid workplace fatigue.
11. As a team manager, I want to view a dashboard summarizing daily activity and readiness scores of multiple employees so that I can get a quick, high-level overview of the team's condition.
12. As a team manager, I want to receive an alert when a team member’s key biometric data, like resting heart rate, deviates significantly from their established baseline so that I can proactively check in on their health and recovery.
13. As a team manager, I want to generate and export data (csv) so that I can analyze all the workers with other tools like a spreadsheet.
14. As a user, I want to sync my watch data with an Android phone's native health platform (Google Fit) so that all my health information is consolidated in one place.
15. As a worker with specialized tools, I want to connect external Bluetooth sensors (like a chest strap heart rate monitor) so that I can capture more accurate and diverse data than the wrist-based sensor alone can provide.
16. As a user, I want the app to track and summarize my activity even when it does not have an internet connection so that I can work on a site without Internet.